Exercise 1.1: Evaluating Model Performance

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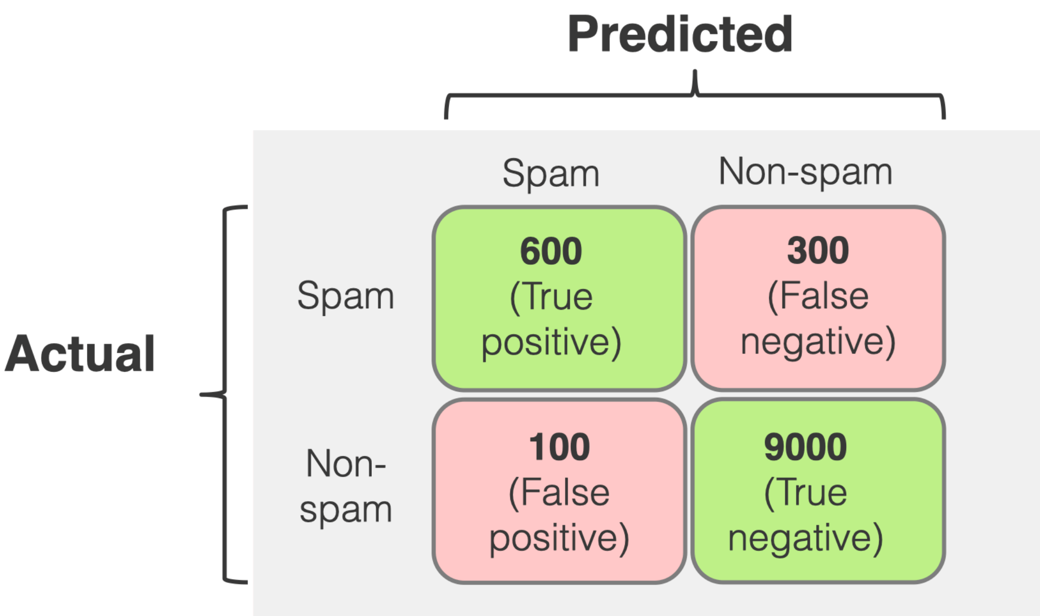
1. Confusion Matrix



Compute the following:

A. Confusion Matrix

- Total = 600 + 300 + 100 + 9000 = 10000
- TP = 600
- FP = 100
- FN = 300
- TN = 9000
- Actual no = 600 + 300 = 900
- Actual yes = 100 + 9000 = 9100
- Predicted no = 600 + 100 = 700
- Predicted yes = 300 + 9000 = 9300

1. Accuracy

   (TP+TN)/total = (600 + 9000) / 10000 = **0.96**

2. Misclassification Rate

(FP+FN)/total = (100 + 300) / 10000 = **0.04**

3. True Positive Rate

   TP/actual yes = 600 / 9100 = **0.0659**

4. False Positive Rate

   FP/actual no = 100 / 9100 = **0.0110**

5. True Negative Rate

   TN/actual no = 9000 / 900 = **10**

6. Precision

   TP/predicted yes = 600 / 9300 = **0.065**

7. Prevalence

   actual yes/total = 9100 / 10000 = **0.91**

B. Classification Report

1. Precision

   TP/(TP + FP) = 600 / (600 + 100) = **0.857**

2. Recall

   TP/(TP+FN) = 600 / (600 + 300) = **0.667**

3. F1 Score

   2 * (Recall * Precision) / (Recall + Precision)

   2 * (0.667 * 0.857) / (0.667 + 0.857) = **0.7502**

4. Support

   TPR = TP/(TP + FN) = 600 / (600 + 300) = **0.667**

   FPR = FP/(FP + TN) = 100 / (100 + 9000) = **0.011**